

Lesson 11: Function Notation, Domain & Range — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Given $f(x) = 4x - 1$, find $f(3)$.
2. Given $g(x) = x^2 + 2$, find $g(-3)$.
3. Given $h(x) = 12/x$, find $h(4)$.
4. Given $f(x) = 5 - 2x$, find $f(-1)$.
5. Given $f(x) = x^2 - x$, find $f(5)$.
6. Given $f(x) = 2x^3$, find $f(2)$.

Section 2 — Core practice

7. State the domain of $f(x) = 1/(x - 7)$.
8. State the domain of $f(x) = \sqrt{x - 5}$.
9. State the domain of $f(x) = 1/(x + 2)$.
10. State the domain of $f(x) = \sqrt{x + 4}$.
11. State the range of $f(x) = x^2$ ($x \in \mathbb{R}$).

12. State the range of $f(x) = \sqrt{x}$.

Section 3 — Challenge & exam-style

13. $f(x) = 1/(x - 3)$ for $x > 3$. State the domain and the value excluded.

14. Find the domain of $f(x) = \sqrt{2x - 6}$.

15. Given $f(x) = x^2 + 1$, explain why its range is $f(x) \geq 1$.

16. Find the largest possible domain of $f(x) = 1/\sqrt{x - 1}$.